School of Computing and Information Technology

CSIT321 - Project

User Manual

Project Topic: CSIT-25-S3-16

(Interactive Website for Real Estate Price Prediction in Singapore)

Group Number: FYP-25-S3-19

Assessor: Premarajan

Supervisor: Terence Chew

Name	UOW ID	SIM Email

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Revision Date	Description	Section Affected	Changes Made by	Version after Revision
6/9	Updated introduction, initial installation instructions, key features	1, 2, 3	Adi	1.1
7/9	Updated initial GUIS 4.1 - 4.6	4	All	1.2
10/9	Updated initial GUIS 4.7 - 4.16	4	All	1.3
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12/9	Edited key features	3	Mandy	1.5

Table of Contents

Document Control....2
(1) Introduction....5
(1.1) What this manual covers....5
(1.2) What we assume about the readers....5
(1.3) Scope and Purpose....5
(2) The Initial Installation Instructions....6
(2.1) Prerequisites....6
(2.2) Backend Set Up....7
(2.3) Frontend Set up....8
(3) Key Features of Our Real Estate Price Prediction Platform in Singapore....9
(4) Initial GUIS of Our Project....9
(4.1) Landing Page....9
(4.2) Sign-up Page....10
(4.2.1) Email Verification....10
(4.2.2) Creating A Password For an Account....11
(4.2.3) Terms and Conditions....11
(4.2.4) Plans & Pricing....12
(4.2.5) Additional User Information for Agent Plan....12
(4.2.6) Payment Section....13
(4.3) Log-in Page....13
(4.4) Log-in Page Wrong Details....14
(4.5) Recovery Page....14
(4.6) Registered Free User Dashboard....15
(4.7) Registered Premium User Dashboard....16
(4.8) Price Prediction Search Page - Registered Free Users and Registered Premium Users....16
(4.8.1) Price Prediction Page - Registered Free Users....17
(4.9) Compare Prediction Search Page - Registered Premium Users....18
(4.9.1) Compare Prediction Page - Registered Premium Users....19
(4.10) Property Listings - Registered Free Users and Registered Premium Users.. 20
(4.11) Bookmarks Page - Free Registered Users....21
(4.11.1) Bookmarks Page - Registered Premium Users....21
(4.12) User Profile - Registered Free Users and Registered Premium Users....22
(4.13) Property Agent - Dashboard....22
(4.13.1) Property Agent - Manage Listings....23
(4.13.2) Property Agent - Manage Regions....24
(4.13.3) Property Agent User Profile....25

(4.14) System Admin Dashboard....26
(4.14.1) User Accounts - System Admin....27
(4.14.2) User Details Under User Accounts....28
(4.14.3) Feedback Management - System Admin....29
(4.14.4) Content Management Under System Admin....30
(4.15) Support Section....31
(4.16) Feedback....32

(1) Introduction

This Preliminary User Manual is written to help a first-time user get our Real Estate Price Prediction (Singapore) web app up and running, and then confidently navigate the very first screens. We're keeping things practical and lightweight so users can go from install → first run → first prediction without digging into advanced features.

(1.1) What this manual covers

- Initial installation on Windows and Mac (frontend + backend).
- First-time configuration (using a simple test setup).
- A quick tour of the initial GUIs: Home and Landing Page, Sign In/Sign Up Page, Price Prediction and Results.

(1.2) What we assume about the readers

- Basic familiarity with opening a terminal and running simple commands.
- Readers know how to copy a project from GitHub and open it in VSCode.

(1.3) Scope and Purpose

- This project is a Real Estate Price Prediction (Singapore) web app. Our goal is to help users get a quick, data-driven estimate of a property's value so they can make smarter decisions before they spend time on viewings or negotiations.
- This Preliminary User Manual only covers initial installation, initial configuration, and the initial GUIs of our project. It's meant to help a new user get from zero to a first successful prediction with minimal friction.

(2) The Initial Installation Instructions

(2.1) Prerequisites

System Requirements

- Python 3.10+ (for Flask backend)
- Node.js 16+ (for React frontend)
- PostgreSQL 12+ with PostGIS extension
- Git (to clone the repository)

Backend Prerequisites

- Python 3.10+ with pip installed
- PostgreSQL installed and running
- PostGIS extension for PostgreSQL
- Python packages (install with pip install -r requirements.txt):
  - Flask 3.0.0
  - Flask-CORS 4.0.0
  - Flask-SQLAlchemy 3.1.1
  - psycopg2-binary 2.9.9 (Windows) or psycopg2 2.9.9 (macOS/Linux)
  - GeoAlchemy2 0.14.2
  - Shapely 2.0.2
  - PyJWT 2.8.0
  - Flask-Mail 0.9.1
  - python-dotenv 1.0.0

Frontend Prerequisites

- Node.js 16+ (comes with npm)
- React 19.1.1 and dependencies (install with npm install)
- React Router DOM 7.8.0 for navigation
- Leaflet 1.9.4 for maps
- React Scripts 5.0.1 for build tools

(2.2) Backend Set Up

Do these in a terminal inside the /backend folder.

Install Python Dependencies

Windows (PowerShell):

- cd backend
- python -m venv .venv
- .venv\Scripts\activate
- pip install -r requirements.txt

MacOS:

- cd backend
- python3 -m venv .venv
- source .venv/bin/activate
- pip install -r requirements.txt

Database setup (PostgreSQL + PostGIS)

Windows (PowerShell):

- psql -U postgres
- CREATE DATABASE fyp_app;
- psql -U postgres -d fyp_app -c "CREATE EXTENSION postgis;"
- psql -U postgres -d fyp_app -f database_fyp.sql

MacOS:

- brew services start postgresql@14
- sudo -u postgres psql or, if your user already has access:
  - psql -U your_mac_username
- psql -U postgres -c "CREATE DATABASE fyp_app;"
- psql -U postgres -d fyp_app -c "CREATE EXTENSION postgis;"
- psql -U postgres -d fyp_app -f database_fyp.sql

Backend Environment Variables

Both Window and MacOs:

- DATABASE_URL=postgresql://username:password@localhost:5432/fyp_app
  - Postgres user/password differs, update DATABASE_URL accordingly.
- SECRET_KEY=your-secret-key-here
- FLASK_ENV=development

Running the Backend Server

Both Window and MacOs:

- export FLASK_APP=app.py
- export FLASK_ENV=development
- flask run

(2.3) Frontend Set up

Frontend Environment Variables

Both Window and MacOs:

- REACT_APP_BACKEND_ORIGIN=http://localhost:5000
- REACT_APP_GOOGLE_MAPS_API_KEY=your-google-maps-key

Both Window and MacOs:

- cd frontend
- npm install
- npm start

(3) Key Features of Our Real Estate Price Prediction Platform in Singapore

- **Instant Property Valuation:** Users can enter details such as property type, floor level and unit size to get an instant and personalized price prediction.
- **Covers Commercial & Industrial Properties:** Unlike most tools that only focus on HDBs and condos, our platform is designed to support commercial and industrial properties, which are often underserved in Singapore .
- **Interactive Maps & Geospatial Insights:** The website uses map data to show nearby amenities like MRT stations and schools. This helps users see how location factors affect property value .
- **Different User Roles:**
  - Unregistered users
  - Registered free users
  - Premium users
  - Agents
  - System Admin

(4) Initial GUIS of Our Project

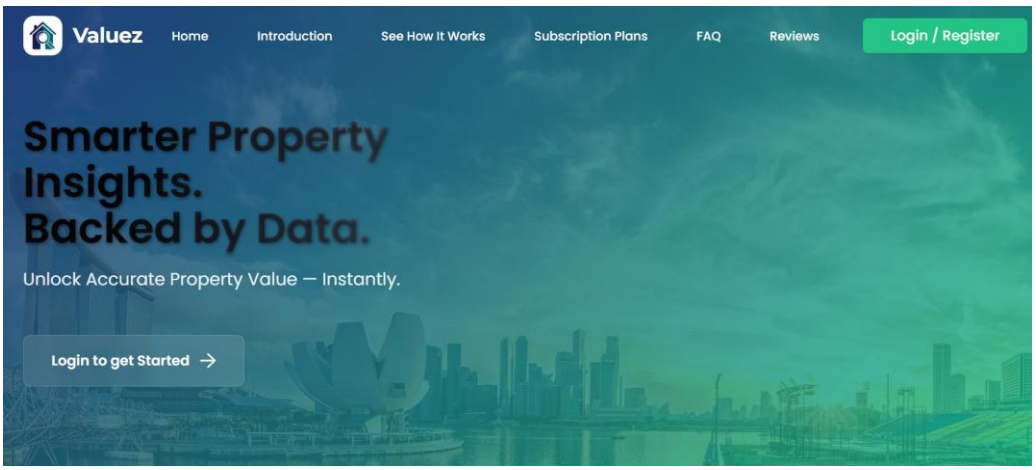
(4.1) Landing Page

This is the first page users see when visiting Valuez.

At the top, there's a menu to explore sections like Introduction, See How It Works, Subscription Plans, FAQ, and Reviews.

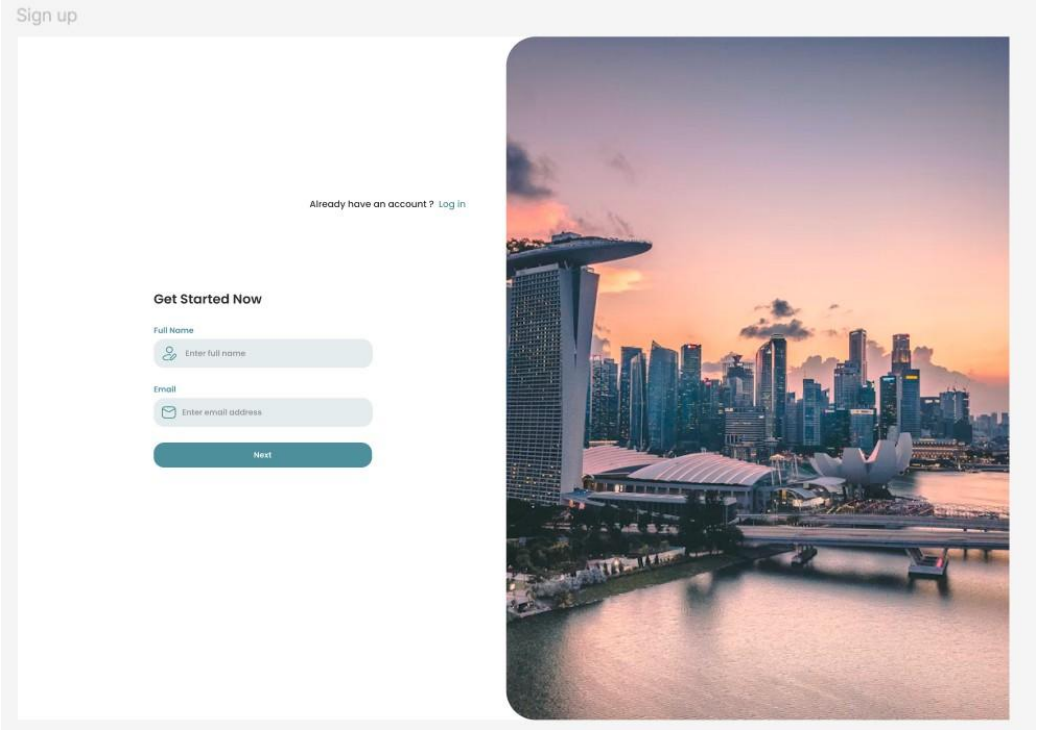
On the top right, a "Login/Register" button lets users sign in or create an account.

A "Login to get Started" button guides users straight into the website.



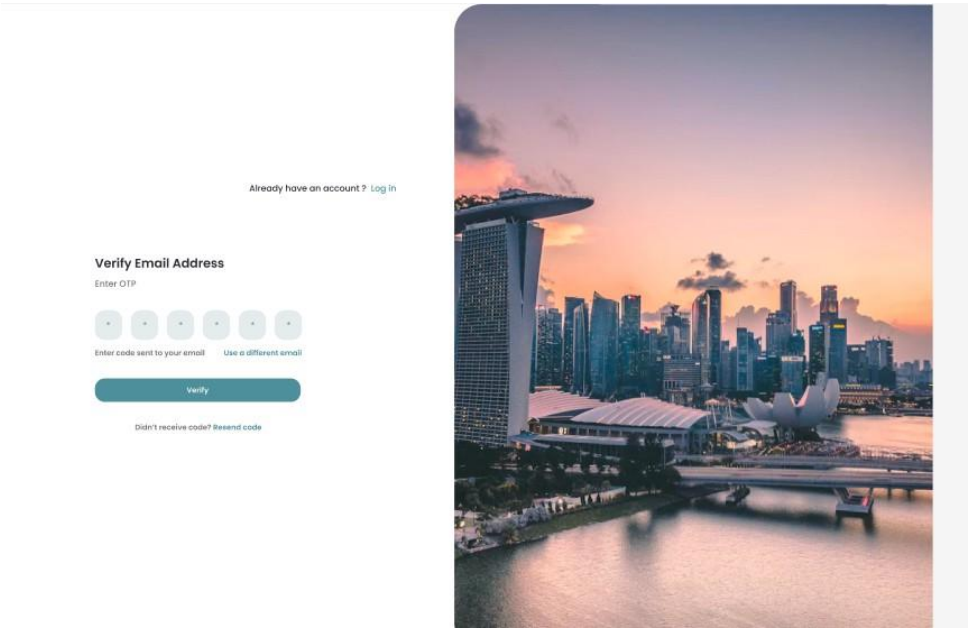
(4.2) Sign-up Page

This page is a sign up page, the user must fill in their full name and email address then the backend will send the verification code to the email address.



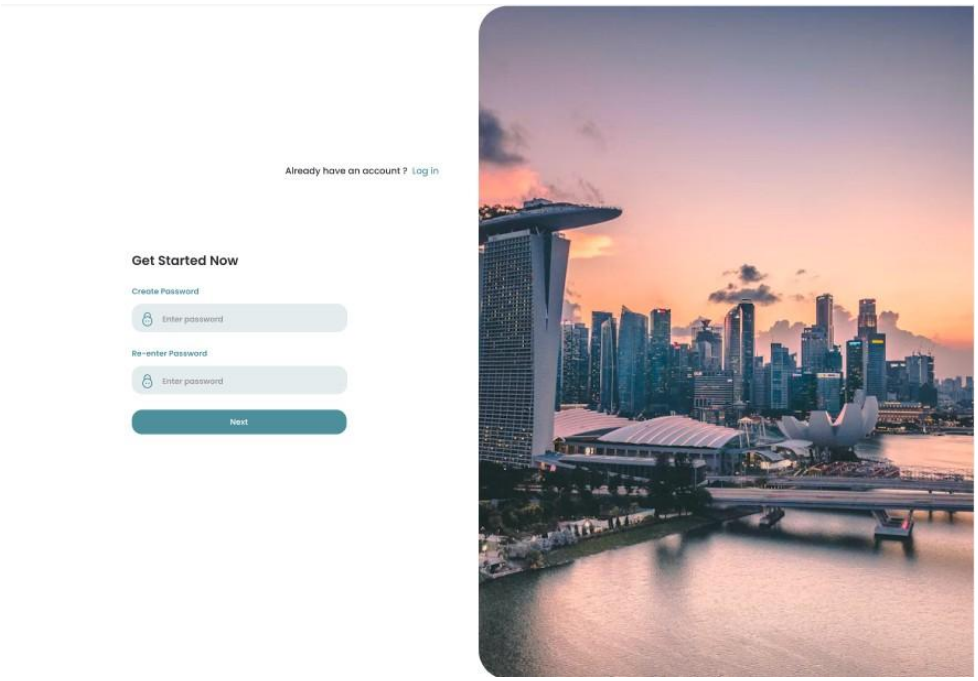
(4.2.1) Email Verification

This page is where the users must type in the verification otp code to continue with the sign up process.



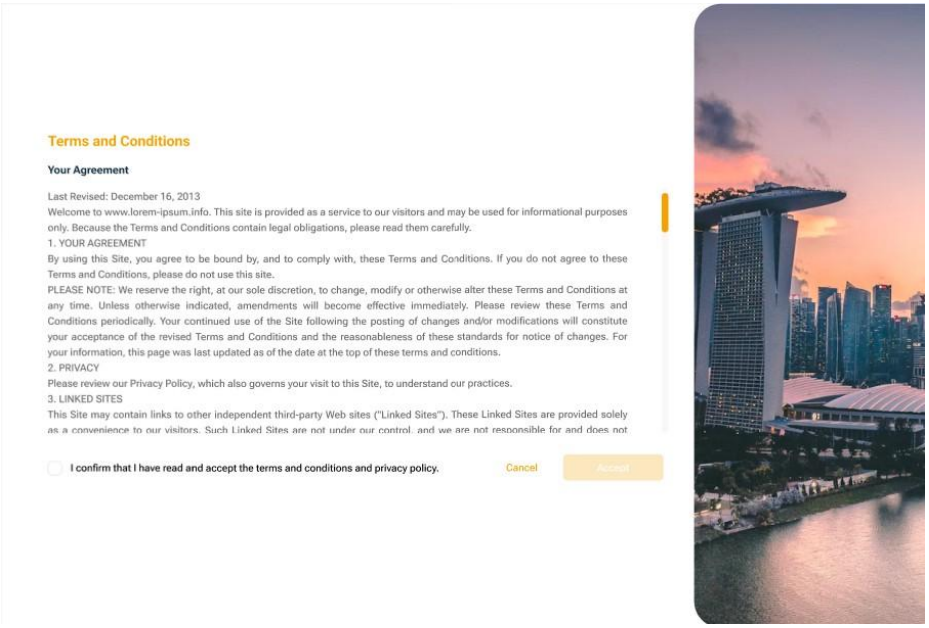
(4.2.2) Creating A Password For an Account

This page is where the users create and confirm their password for the account.



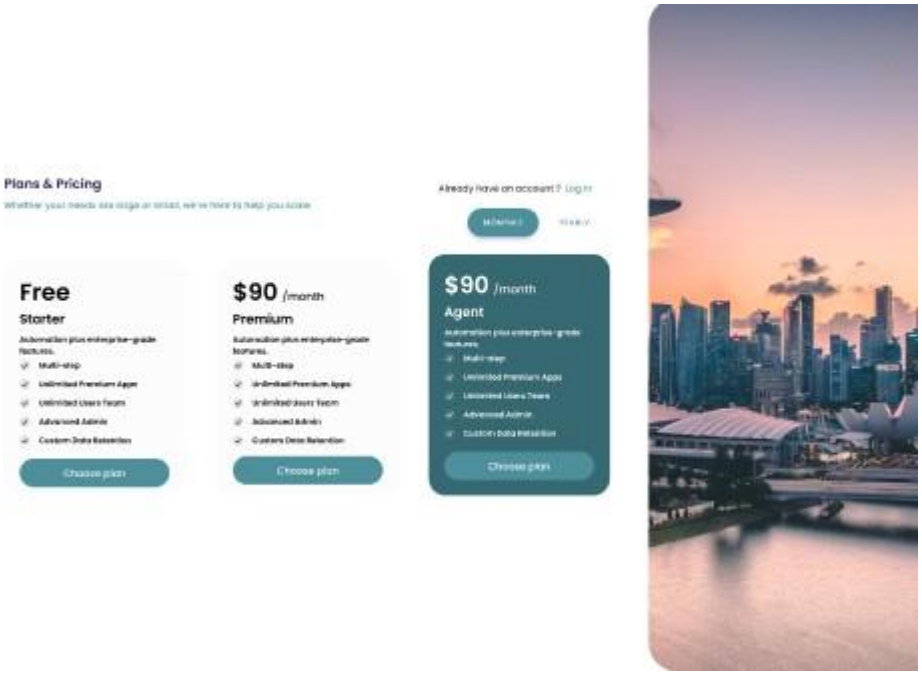
(4.2.3) Terms and Conditions

Users need to read and accept our terms and conditions to continue the sign up process.



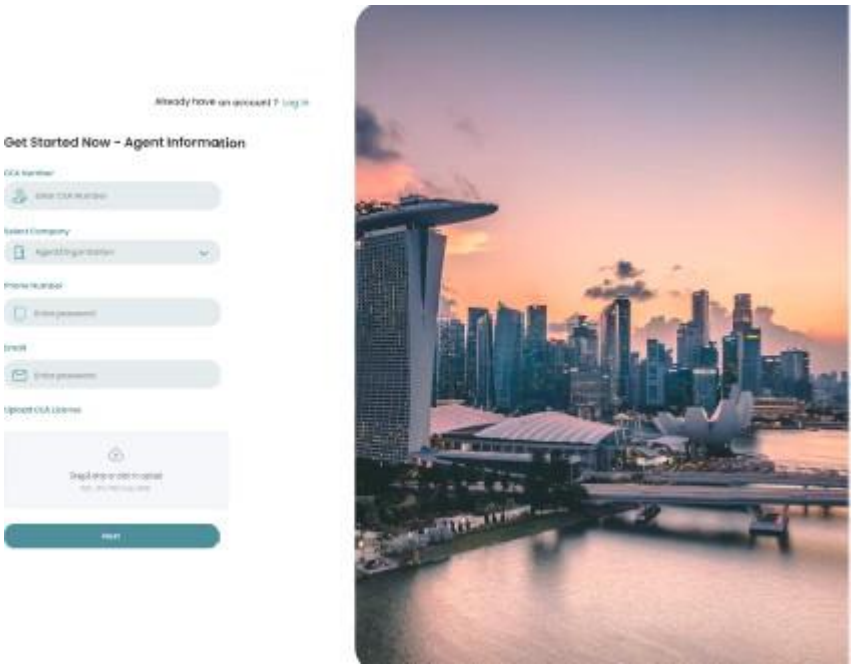
(4.2.4) Plans & Pricing

Users can choose the plans to sign up. (free, premium, agents)



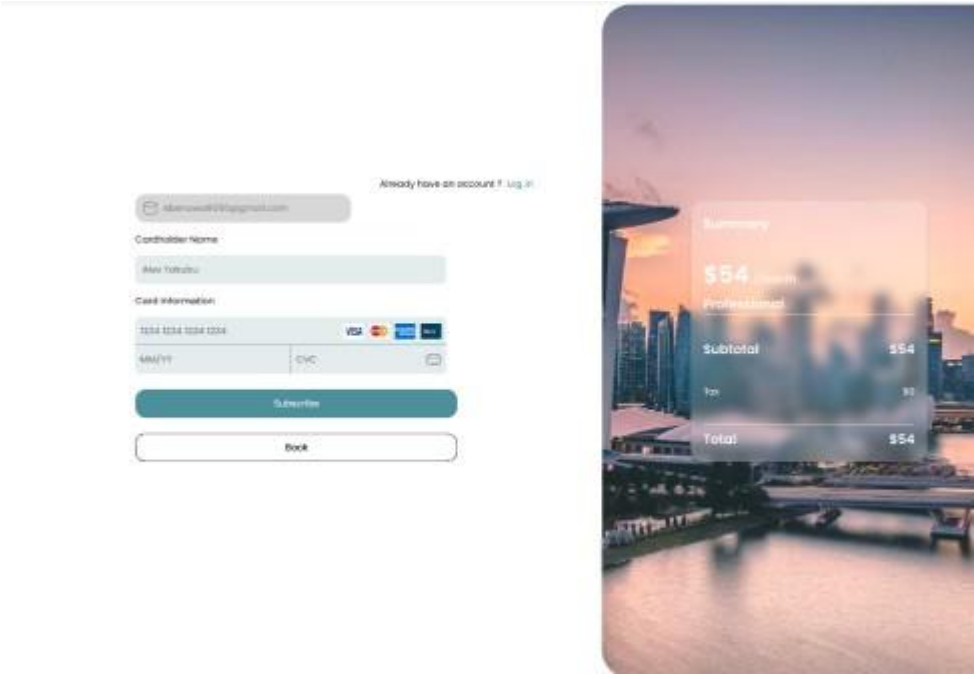
(4.2.5) Additional User Information for Agent Plan

Additional information details for the agent to fill up before proceeding to the payment page.



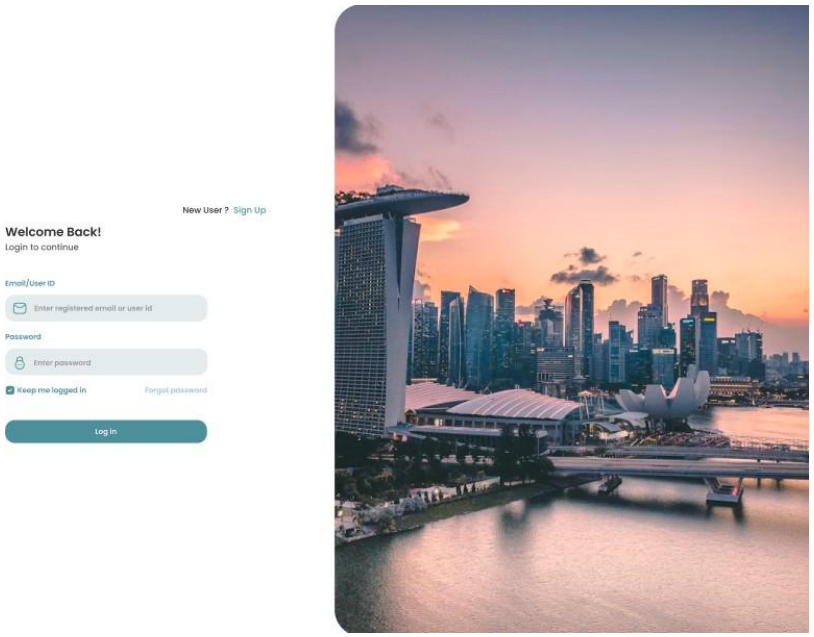
(4.2.6) Payment Section

Payment page if the user clicks the premium plan, or user clicks the agent plan and fills up the additional information required for agents.



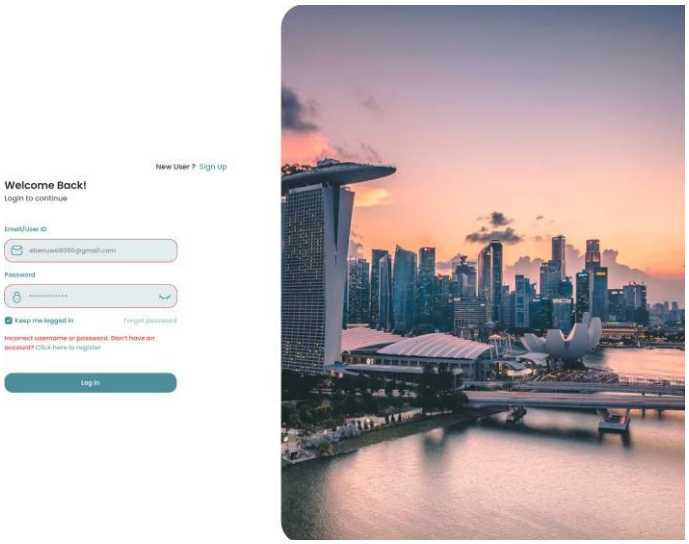
(4.3) Log-in Page

This is the log-in page for all users.



(4.4) Log-in Page Wrong Details

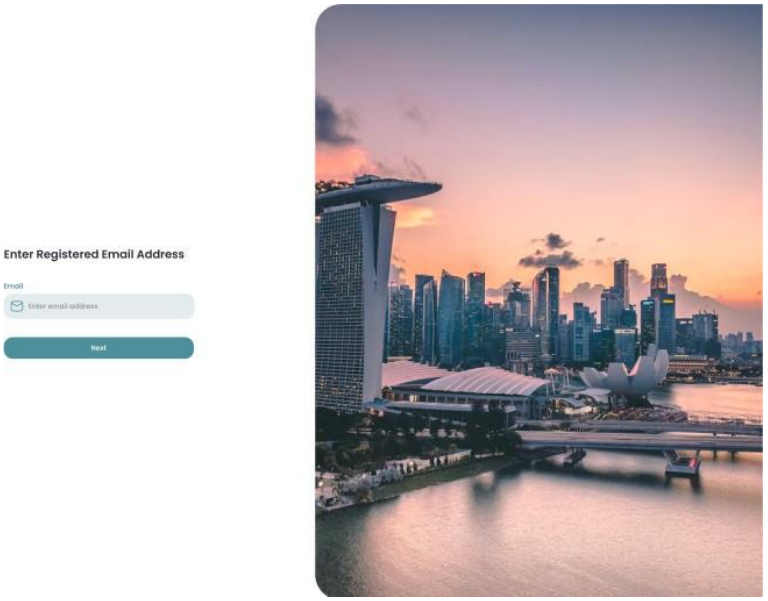
If a user accidentally type the wrong user name or password, the website will prompt as follow:



(4.5) Recovery Page

If users forget the password, users can recover the account using the registered e-mail address.The steps are as follow:

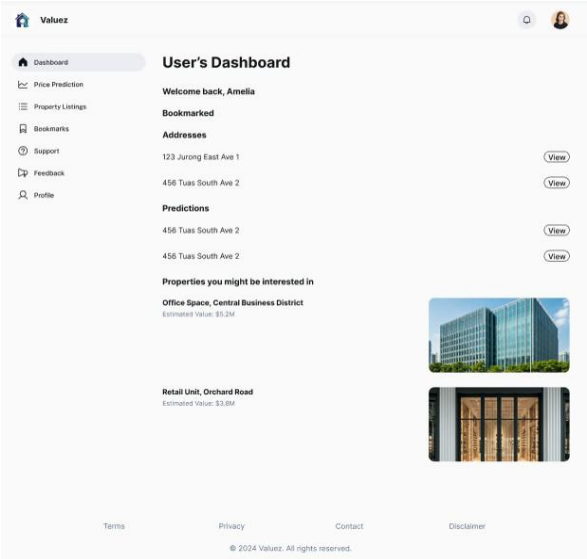
- First step, enter registered email address.



- Users can create a new password.

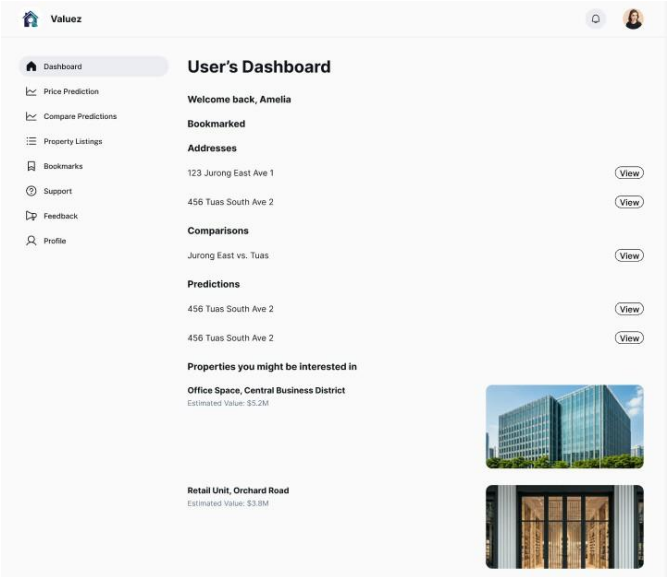
(4.6) Registered Free User Dashboard

If a user log in the app as a registered free user, the user can see as per follow:



(4.7) Registered Premium User Dashboard

The difference between the free users and premium users is the premium user can compare the price predictions. If a user log in as a premium user, the user can see as per follow:

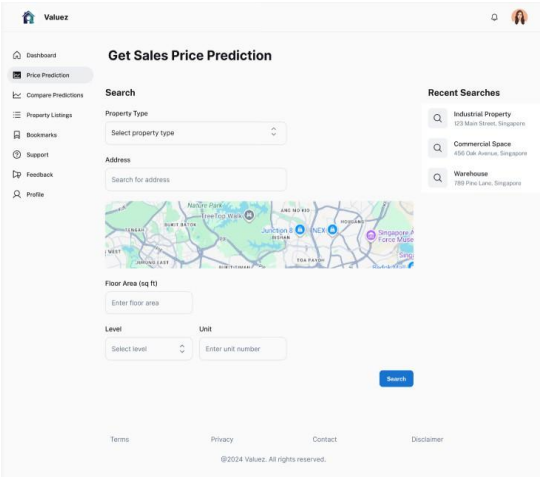


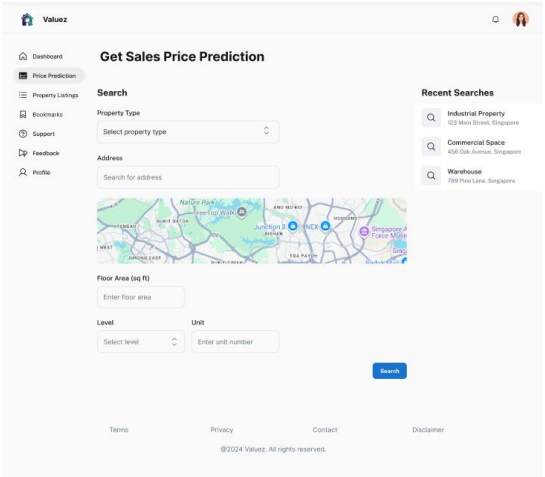
(4.8) Price Prediction Search Page - Registered Free Users and Registered Premium Users

For Free Users and Premium users, the users can compare two properties.

How to use:

- ➢ Select Property Type from the dropdown.
- ➢ Enter Address and choose from the suggestions.
- ➢ Fill Floor Area, Level, and Unit.
- ➢ Hit Search to see prediction.
- ➢ Use Recent Predictions on the right to re-run past comparisons quickly.

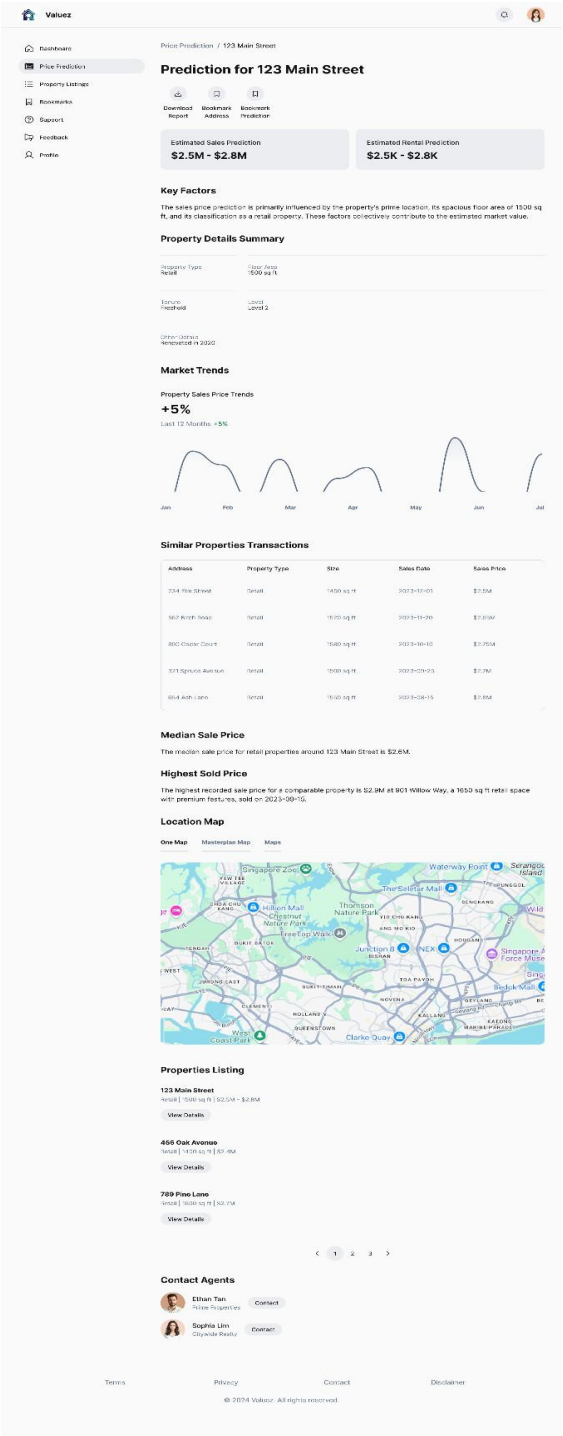




(4.8.1) Price Prediction Page - Registered Free Users

After searching the property information, the user can see the prediction page.

- ➢ The user can see the estimated sale prediction and estimated rental prediction.
- ➢ The user also can see the property details summary and market trends which are valid for the last 12 months.
- ➢ The user also can see the similar properties transactions and the property location on the map.

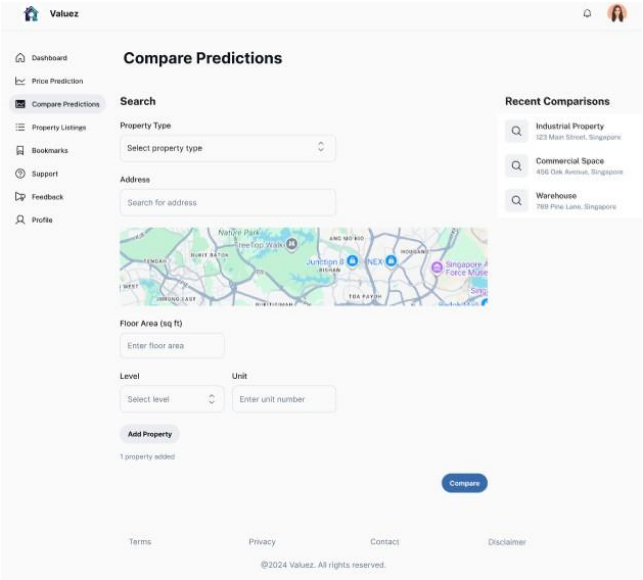


(4.9) Compare Prediction Search Page - Registered Premium Users

For Registered Premium Users, the users can compare the price prediction.

How to use:

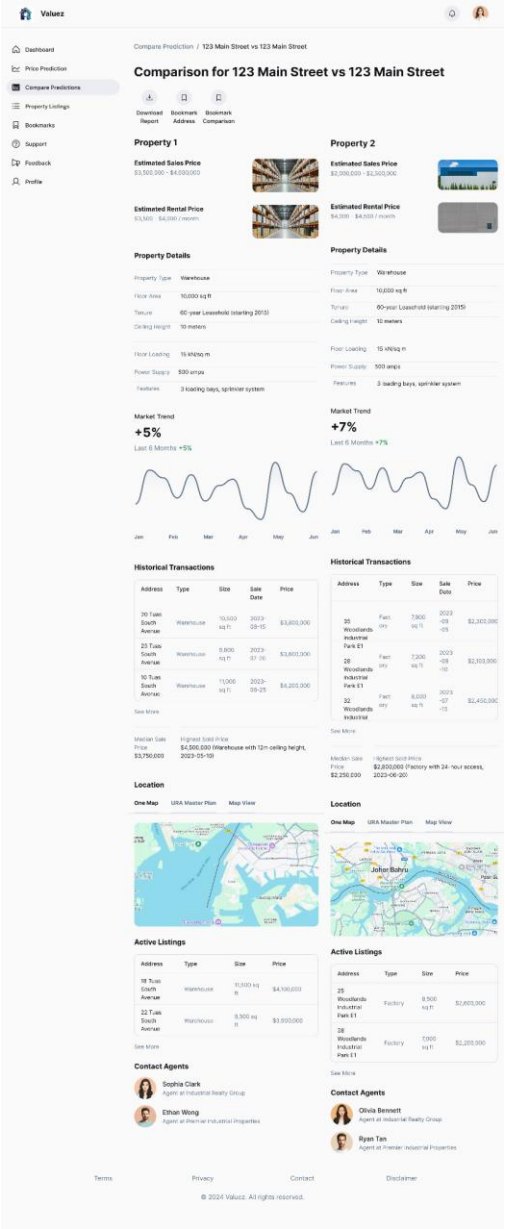
- ➢ Select Property Type from the dropdown.
- ➢ Enter Address and choose from the suggestions.
- ➢ Fill Floor Area, Level, and Unit.
- ➢ Click Add Property and fill in the details for the second property.
- ➢ Hit Compare to see side-by-side predictions.
- ➢ Use Recent Comparisons on the right to re-run past comparisons quickly.



(4.9.1) Compare Prediction Page - Registered Premium Users

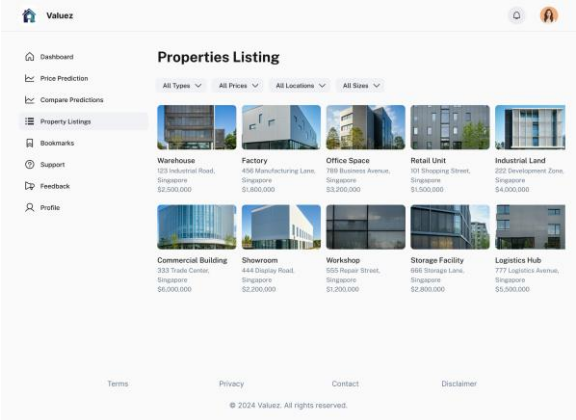
After searching, the user can see the comparison page.

- ➢ Gives a direct property vs property comparison.
    - The user can see the estimated sale prediction and estimated rental prediction for each property.
    - The user also can see the property details summary and market trends which are valid for the last 12 months for each property.
    - The user also can see the similar properties transactions and the property location on the map for each property.

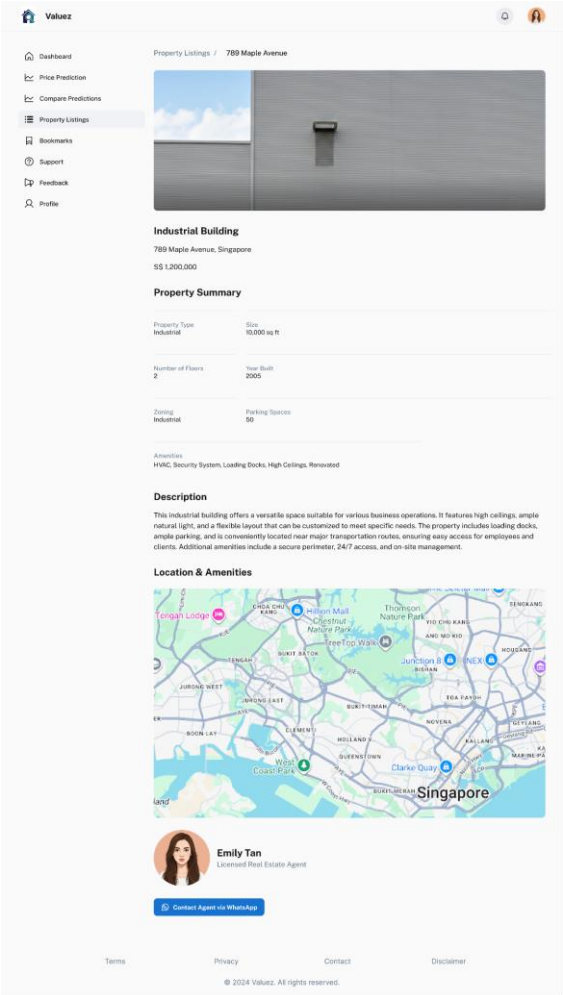


(4.10) Property Listings - Registered Free Users and Registered Premium Users

In the property listing page, the users can see all the listings added by agents on the website.



If users click the individual property listings, the users can see as per follow:

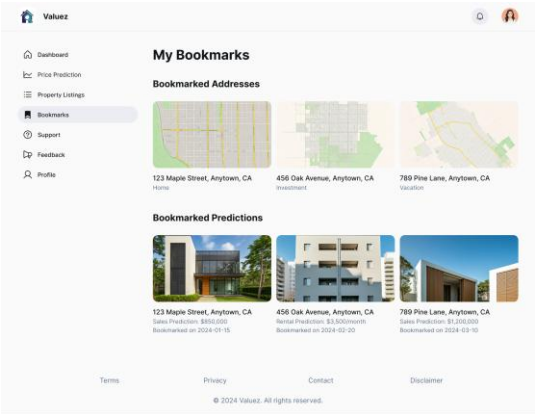


(4.11) Bookmarks Page - Free Registered Users

Free Registered Users can save important property addresses and property details for quick access later by clicking the “Bookmark Address” or “Bookmark Prediction” button in the prediction page.

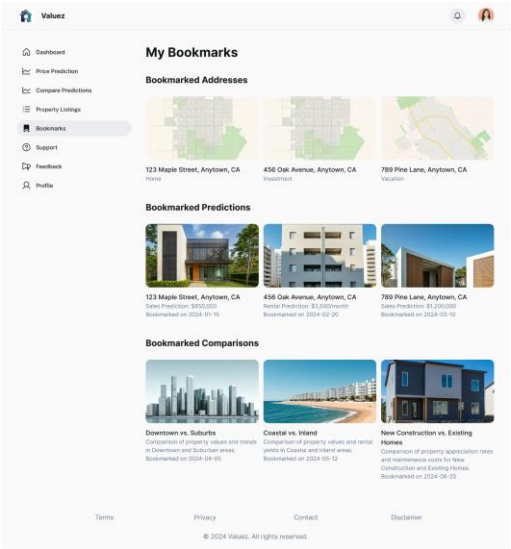
In the Bookmarks Page,

- ➢ Bookmarked Address Section is where all the Bookmark Address are being saved, users can click on the bookmark item for easy searching of the different properties of the same address.
- ➢ Bookmarked Predictions Section is where all the Bookmark Predictions are being saved, users can click on the Bookmark item for predicting the same property at different time periods to analyse or keep track of the changes for that property.



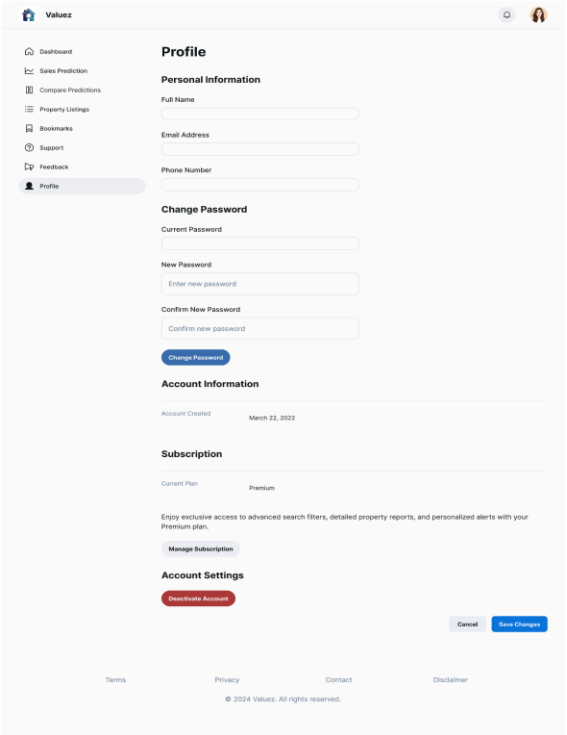
(4.11.1) Bookmarks Page - Registered Premium Users

For the Registered Premium Users Bookmarks Page, users can see the Bookmarked Comparisons Section where all the Bookmark Comparisons are being saved, users can click on the Bookmark item for comparing the two properties at different time periods to analyse or keep track of the changes for these two properties.



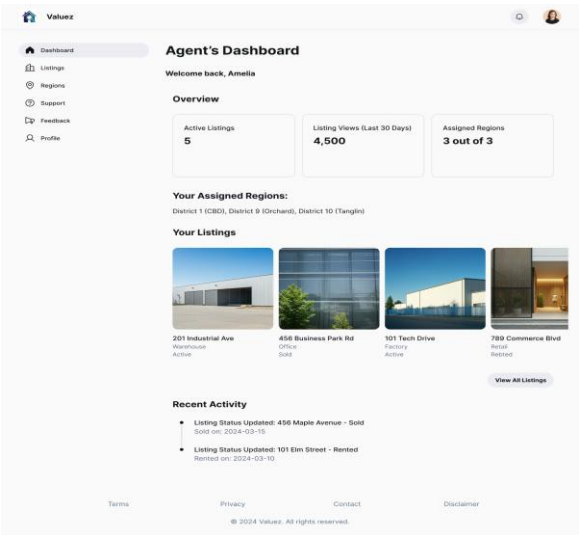
(4.12) User Profile - Registered Free Users and Registered Premium Users

- ➢ Users can edit their personal information like full name, email address and phone number.
- ➢ Users can also change their password and update on this page.

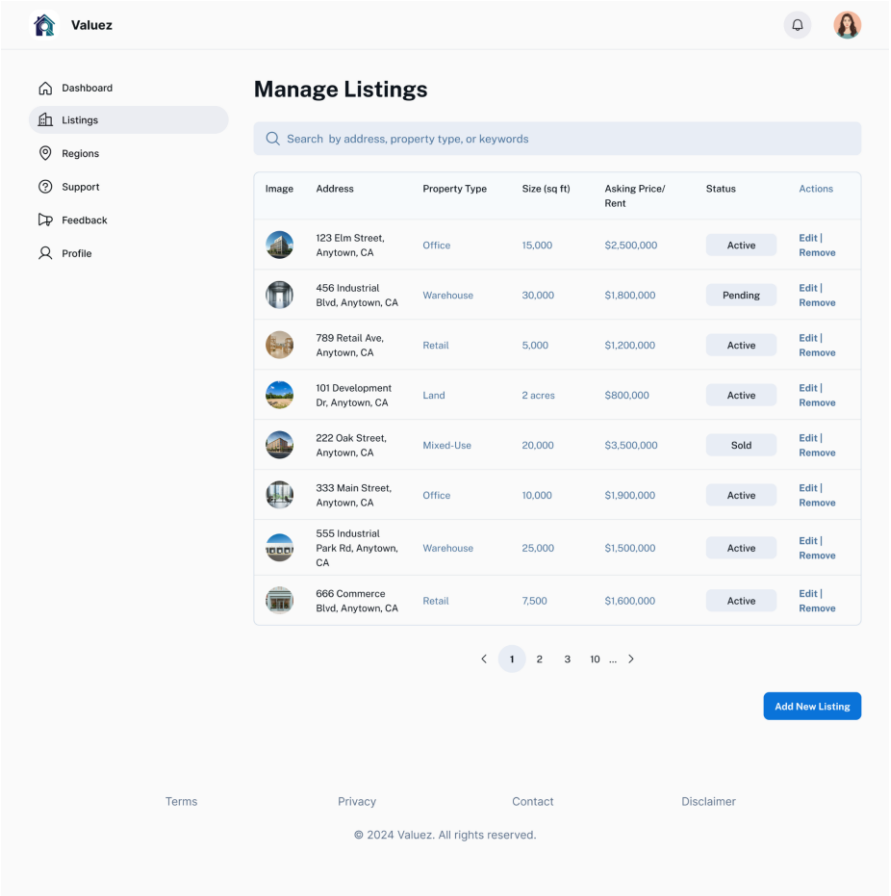


(4.13) Property Agent - Dashboard

If a user log into the website as a property agent user, the user can see as per follow:

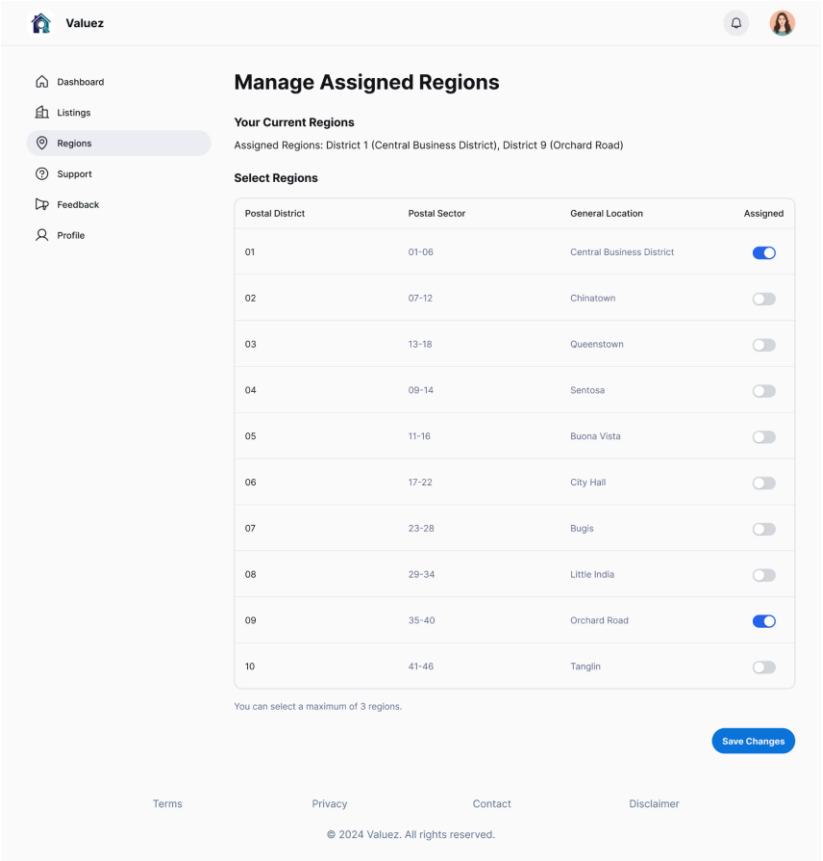


(4.13.1) Property Agent - Manage Listings



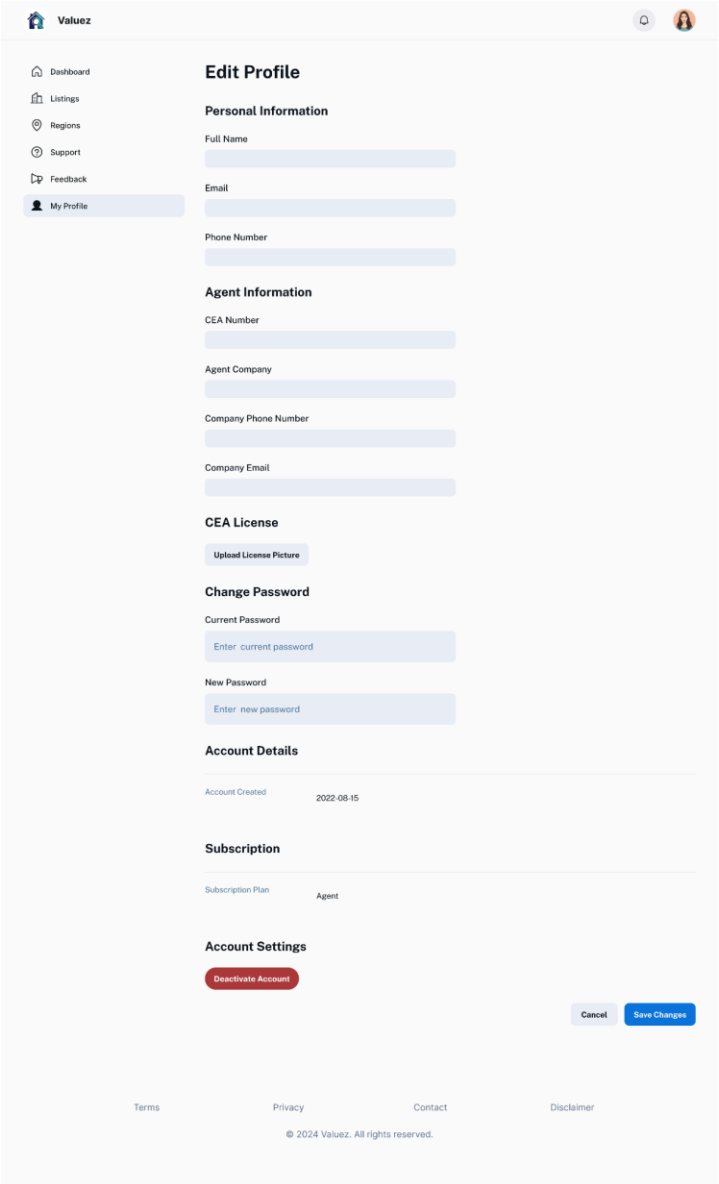
- ➢ Property Agents can see, edit, and control all their property listings.
- ➢ Property Overview Table: Lists each property with its image, address, property type, size, asking price/rent, and status (Active, Pending, Sold).
- ➢ Quick Search: Search listings by address, property type, or keywords.
- ➢ Actions Column: Edit details or remove a listing anytime.
- ➢ Add New Listing Button: Fill in the property listing details to add a new property listing to the website.

(4.13.2) Property Agent - Manage Regions



- This page helps agents choose and manage the regions they're in charge of.
- Current Regions Overview: Shows all the regions already assigned to the agent.
- Region Selection Table: Lists regions by Postal District, Sector, and General Location.
- Toggle switches to assign or unassign regions.
- Assignment Limit: A note shows the max number of regions that can be assigned (e.g., 3).
- Save Changes Button: Save updated assignments with one click.

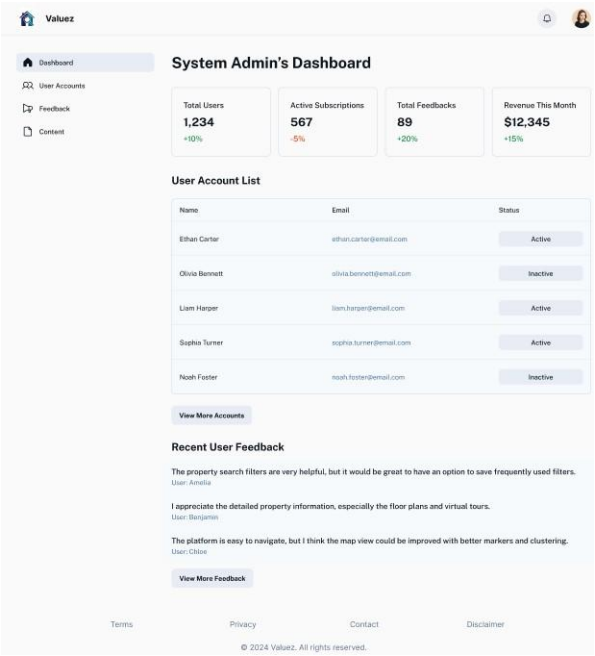
(4.13.3) Property Agent User Profile



- Agent Profile Page is designed for property agents to manage their professional details, CEA license info, and account security.
- Personal Information: Update name, email, and phone number.
- Agent Information: Fill in CEA Number, company name, and contact details. Upload a CEA license image to verify credentials.
- Password Management: Change your password securely by entering current and new passwords.
- Account Details: Displays when your account was created.
- Subscription Plan: Shows your plan type (Agent).
- Deactivate Account: Option to permanently remove your account.

(4.14) System Admin Dashboard

If a user log into the website as a System Admin user, the user can see as per follow:



Analytics Cards:

- ➢ Total Users: Number of registered users.
- ➢ Active Subscriptions: Count of active subscriptions.
- ➢ Total Feedbacks: Shows engagement from user feedback submissions.
- ➢ Monthly Revenue: Tracks total earnings for the current month.

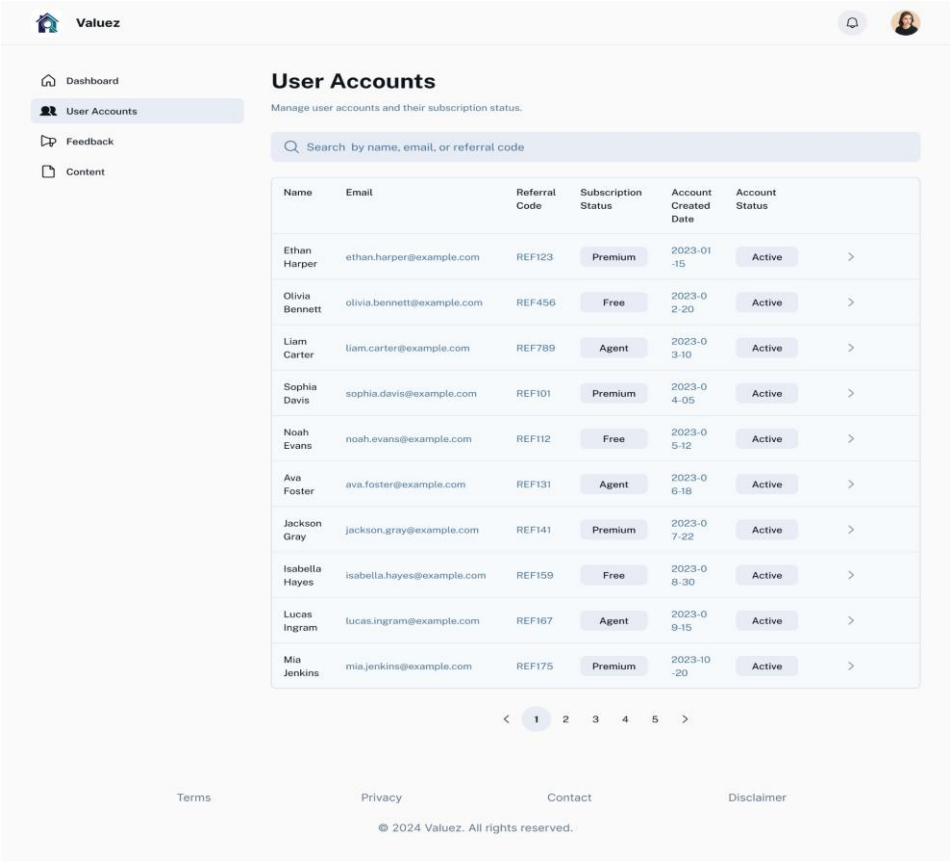
User Account List:

- ➢ Quick view of all users with their account status (Active/Inactive).
- ➢ Access to manage or update user records.

Recent User Feedback:

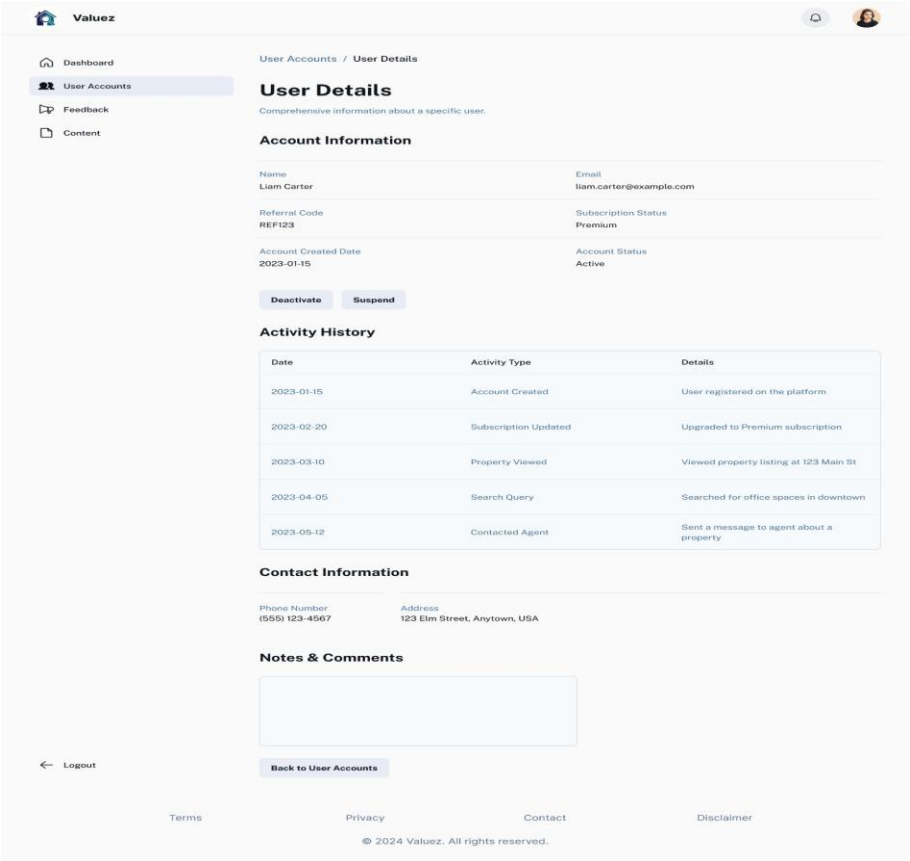
- ➢ Shows latest reviews from users.
- ➢ Helps identify areas of improvement.

(4.14.1) User Accounts - System Admin



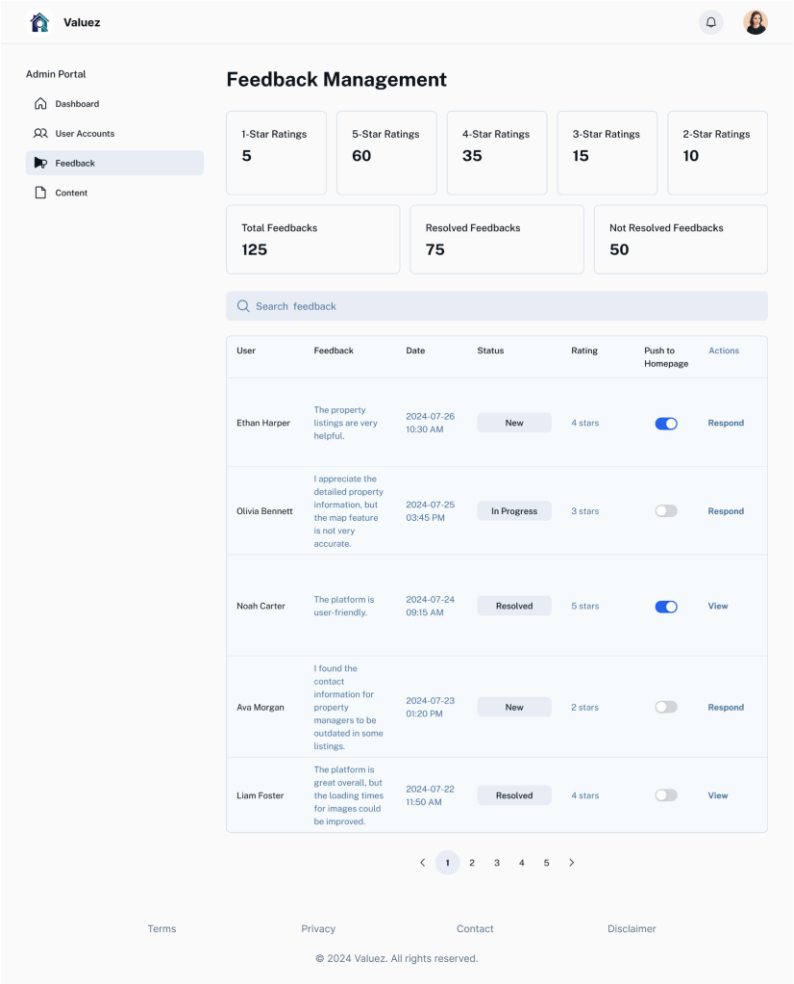
- This page allows System Admin to view and manage every user's details, subscription plan, and account status.
- User List Table: Shows each user's name, email, referral code, subscription type (Free, Premium, Agent), and account status. Displays the date the account was created.
- Search Bar: Search users by name or email.
- Account Status Control: Quickly check who is Active or Inactive.

(4.14.2) User Details Under User Accounts



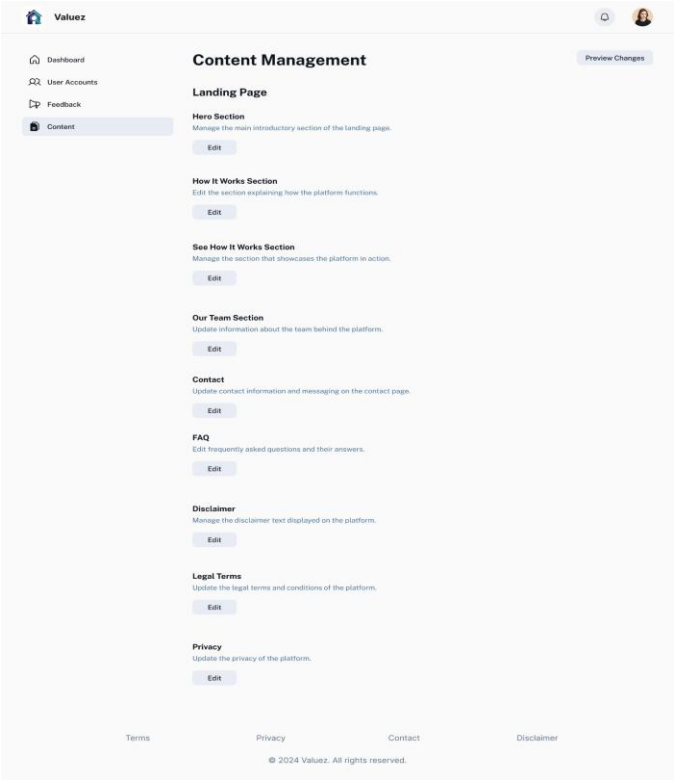
- This page lets System Admin view and manage detailed information about a specific user.
- Account Information: Shows user's name, email, referral code, subscription status, and account status.
- Account Controls: Admins can Deactivate or Suspend the user account.
- Activity History: Timeline of user actions
- Contact Information: Displays user's phone number and address for verification or support.
- Notes & Comments: A section for admins to leave internal notes or observations.

(4.14.3) Feedback Management - System Admin



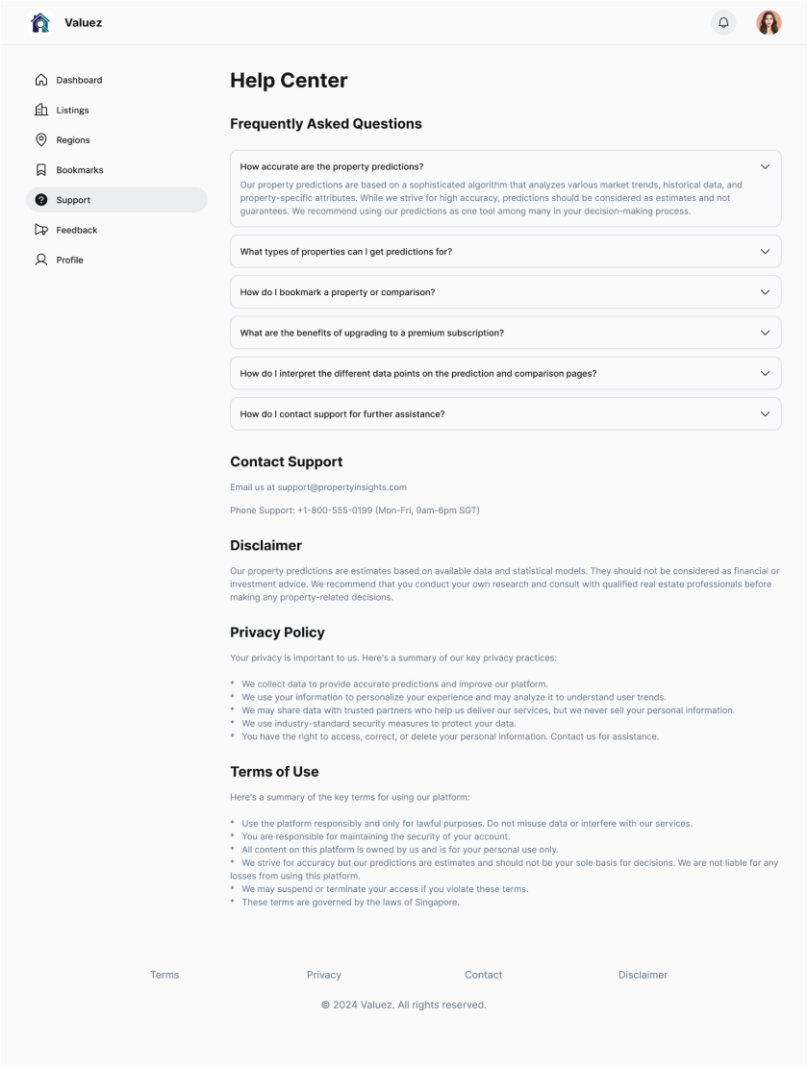
- This is the Feedback Management page for System Admin. It's designed to handle all user feedback and ratings in one place.
- This page allows System Admin to view, track, and respond to all user feedback, helping to improve the platform.
- Feedback Summary Cards: Shows counts of ratings by stars (1 to 5 stars). Displays total feedback, resolved feedback, and unresolved feedback numbers.
- Feedback Table: Lists feedback by user, date, rating, and status (New, In Progress, Resolved). Shows ratings for each review and option for system admin to quickly push positive reviews to the homepage or other section for other users to see.
- Search Bar: Quickly search feedback by keywords or user names.
- Actions Column: Admins can respond to or view feedback details.

(4.14.4) Content Management Under System Admin



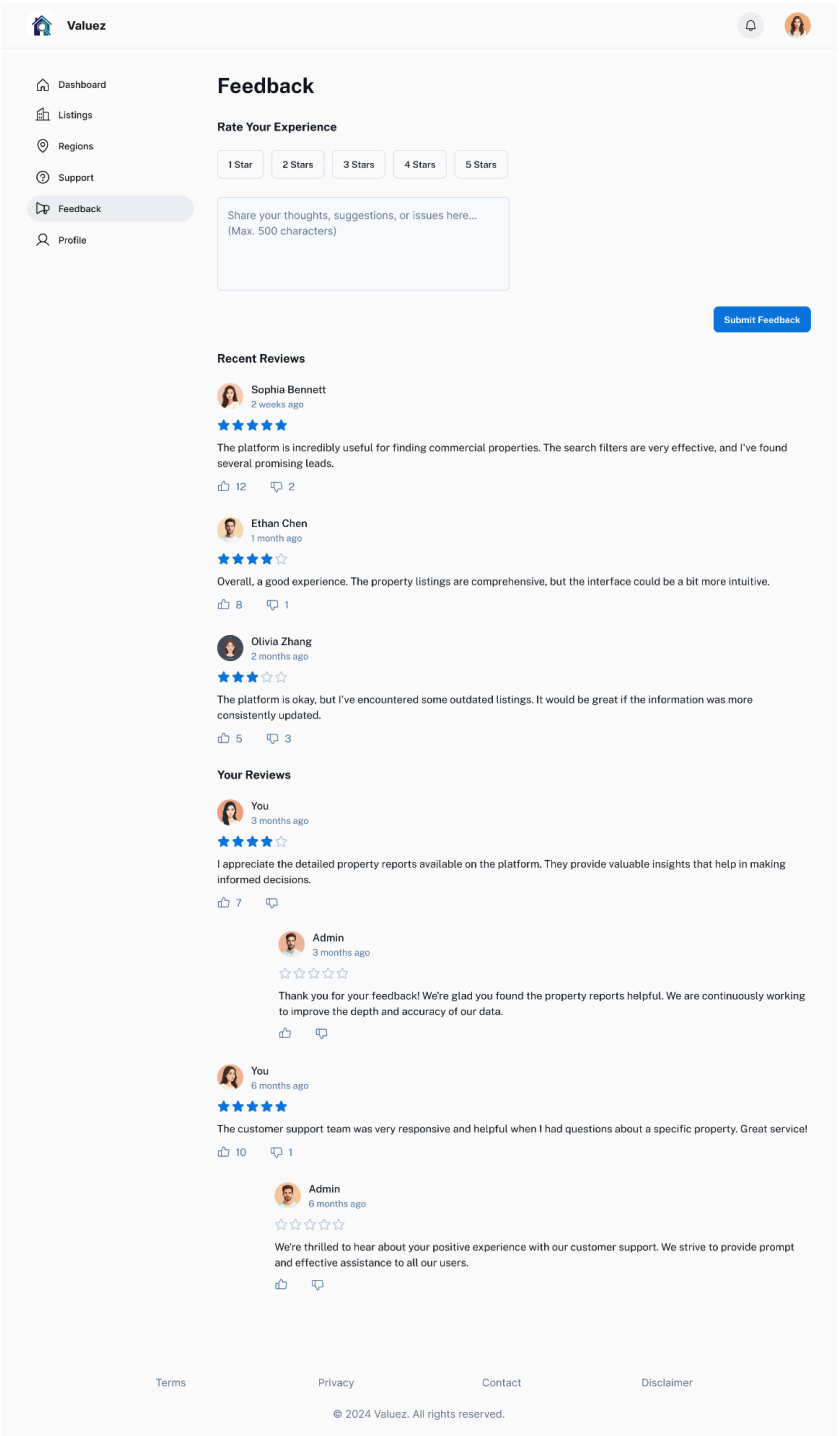
- This page allows System Administrators to update and manage the text, visuals, and messaging across the platform's landing page and essential info sections.
- Hero Section: Edit the main introduction message users see when visiting the platform.
- How It Works Section: Edit short descriptions of how your platform functions.
- See How It Works Section: Edit the property details for the sample predictions.
- Our Team Section: Update information about the development team.
- Contact Section: Edit support email, address, and contact page information.
- FAQ Section: Manage frequently asked questions and their answers.
- Disclaimer: Update platform disclaimer text for users.
- Legal Terms: Maintain terms and conditions.
- Privacy Section: Manage privacy policy and data-handling disclosures.
- Preview Changes Button: Quickly review edits that have been saved.

(4.15) Support Section



- This page is users' go-to guide for answers to common questions, support contacts, platform disclaimers, privacy practices, and terms of use.

(4.16) Feedback



- This page lets users rate their experience, write reviews, and view other users' feedback to improve the platform.
- Rate Your Experience: Choose a star rating (1 to 5 stars) for your overall experience.

- Write Feedback: Add comments, suggestions, or issues (up to 500 characters). Helps the team improve features based on your input.
- Submit Feedback Button: Click to send your feedback instantly.
- Recent Reviews Section: See what other users are saying about the platform.Includes likes, comments, and review dates.
- Your Reviews Section: Shows all feedback you've previously submitted for reference.